tively. If provided, *locals* can be any mapping object. Remember that at the module level, globals and locals are the same dictionary.

> **Note:** When `exec` gets two separate objects as *globals* and *locals*, the code will be executed as if it were embedded in a class definition. This means functions and classes defined in the executed code will not be able to access variables assigned at the top level (as the "top level" variables are treated as class variables in a class definition).

If the *globals* dictionary does not contain a value for the key `__builtins__`, a reference to the dictionary of the built-in module `builtins` is inserted under that key. Overriding `__builtins__` can be used to restrict or change the available names, but this is **not** a security mechanism: the executed code can still access all builtins.

The *closure* argument specifies a closure–a tuple of cellvars. It's only valid when the *object* is a code object containing free (closure) variables. The length of the tuple must exactly match the length of the code object's `co_freevars` attribute.

Raises an auditing event `exec` with the code object as the argument. Code compilation events may also be raised.

> **Note:** The built-in functions `globals()` and `locals()` return the current global and local namespace, respectively, which may be useful to pass around for use as the second and third argument to `exec()`.

> **Note:** The default *locals* act as described for function `locals()` below. Pass an explicit *locals* dictionary if you need to see effects of the code on *locals* after function `exec()` returns.

Changed in version 3.11: Added the *closure* parameter.

Changed in version 3.13: The *globals* and *locals* arguments can now be passed as keywords.

Changed in version 3.13: The semantics of the default *locals* namespace have been adjusted as described for the `locals()` builtin.

filter(*function, iterable, /*)

Construct an iterator from those elements of *iterable* for which *function* is true. *iterable* may be either a sequence, a container which supports iteration, or an iterator. If *function* is `None`, the identity function is assumed, that is, all elements of *iterable* that are false are removed.

Note that `filter(function, iterable)` is equivalent to the generator expression `(item for item in iterable if function(item))` if function is not `None` and `(item for item in iterable if item)` if function is `None`.

See `itertools.filterfalse()` for the complementary function that returns elements of *iterable* for which *function* is false.

class **float**(*number=0.0, /*)
class **float**(*string, /*)

Return a floating-point number constructed from a number or a string.

Examples:

```
>>> float('+1.23')
1.23
>>> float('   -12345\n')
-12345.0
>>> float('1e-003')
0.001
>>> float('+1E6')
1000000.0
>>> float('-Infinity')
-inf
```

If the argument is a string, it should contain a decimal number, optionally preceded by a sign, and optionally embedded in whitespace. The optional sign may be `'+'` or `'-'`; a `'+'` sign has no effect on the value produced. The argument may also be a string representing a NaN (not-a-number), or positive or negative infinity. More precisely, the input must conform to the `floatvalue` production rule in the following grammar, after leading and trailing whitespace characters are removed:

```
sign:          "+" | "-"
infinity:      "Infinity" | "inf"
nan:           "nan"
digit:         <a Unicode decimal digit, i.e. characters in Unicode general category Nd>
digitpart:     digit (["_"] digit)*
number:        [digitpart] "." digitpart | digitpart ["."]
exponent:      ("e" | "E") [sign] digitpart
floatnumber:   number [exponent]
absfloatvalue: floatnumber | infinity | nan
floatvalue:    [sign] absfloatvalue
```

Case is not significant, so, for example, "inf", "Inf", "INFINITY", and "iNfINity" are all acceptable spellings for positive infinity.

Otherwise, if the argument is an integer or a floating-point number, a floating-point number with the same value (within Python's floating-point precision) is returned. If the argument is outside the range of a Python float, an `OverflowError` will be raised.

For a general Python object `x`, `float(x)` delegates to `x.__float__()`. If `__float__()` is not defined then it falls back to `__index__()`.

See also `float.from_number()` which only accepts a numeric argument.

If no argument is given, `0.0` is returned.

The float type is described in Numeric Types — int, float, complex.

Changed in version 3.6: Grouping digits with underscores as in code literals is allowed.

Changed in version 3.7: The parameter is now positional-only.

Changed in version 3.8: Falls back to `__index__()` if `__float__()` is not defined.

format(*value, format_spec='', /*)

Convert a *value* to a "formatted" representation, as controlled by *format_spec*. The interpretation of *format_spec* will depend on the type of the *value* argument; however, there is a standard formatting syntax that is used by most built-in types: Format specification mini-language.

The default *format_spec* is an empty string which usually gives the same effect as calling `str(value)`.

A call to `format(value, format_spec)` is translated to `type(value).__format__(value, format_spec)` which bypasses the instance dictionary when searching for the value's `__format__()` method. A `TypeError` exception is raised if the method search reaches `object` and the *format_spec* is non-empty, or if either the *format_spec* or the return value are not strings.

Changed in version 3.4: `object().__format__(format_spec)` raises `TypeError` if *format_spec* is not an empty string.

class **frozenset**(*iterable=(), /*)

Return a new `frozenset` object, optionally with elements taken from *iterable*. `frozenset` is a built-in class. See also Set Types — set, frozenset for documentation about this class.

For other containers see the built-in `set`, `list`, `tuple`, and `dict` classes, as well as the `collections` module.

getattr(*object, name, /*)
getattr(*object, name, default, /*)

Return the value of the named attribute of *object*. *name* must be a string. If the string is the name of one of the object's attributes, the result is the value of that attribute. For example, `getattr(x, 'foobar')` is equivalent to `x.foobar`. If the named attribute does not exist, *default* is returned if provided, otherwise `AttributeError` is raised. *name* need not be a Python identifier (see `setattr()`).

> **Note:** Since private name mangling happens at compilation time, one must manually mangle a private attribute's (attributes with two leading underscores) name in order to retrieve it with `getattr()`.

globals()

Return the dictionary implementing the current module namespace. For code within functions, this is set when the function is defined and remains the same regardless of where the function is called.

hasattr(*object, name, /*)

The arguments are an object and a string. The result is `True` if the string is the name of one of the object's attributes, `False` if not. (This is implemented by calling `getattr(object, name)` and seeing whether it raises an `AttributeError` or not.)

hash(*object, /*)

Return the hash value of the object (if it has one). Hash values are integers. They are used to quickly compare dictionary keys during a dictionary lookup. Numeric values that compare equal have the same hash value (even if they are of different types, as is the case for 1 and 1.0).

> **Note:** For objects with custom `__hash__()` methods, note that `hash()` truncates the return value based on the bit width of the host machine.

help()
help(*request*)

Invoke the built-in help system. (This function is intended for interactive use.) If no argument is given, the interactive help system starts on the interpreter console. If the argument is a string, then the string is looked up as the name of a module, function, class, method, keyword, or documentation topic, and a help page is printed on the console. If the argument is any other kind of object, a help page on the object is generated.

Note that if a slash(/) appears in the parameter list of a function when invoking `help()`, it means that the parameters prior to the slash are positional-only. For more info, see the FAQ entry on positional-only parameters.

This function is added to the built-in namespace by the `site` module.

Changed in version 3.4: Changes to `pydoc` and `inspect` mean that the reported signatures for callables are now more comprehensive and consistent.

hex(*integer, /*)

Convert an integer number to a lowercase hexadecimal string prefixed with "0x". If *integer* is not a Python `int` object, it has to define an `__index__()` method that returns an integer. Some examples:

```
>>> hex(255)
'0xff'
>>> hex(-42)
'-0x2a'
```

If you want to convert an integer number to an uppercase or lower hexadecimal string with prefix or not, you can use either of the following ways:

```
>>> '%#x' % 255, '%x' % 255, '%X' % 255
('0xff', 'ff', 'FF')
>>> format(255, '#x'), format(255, 'x'), format(255, 'X')
('0xff', 'ff', 'FF')
>>> f'{255:#x}', f'{255:x}', f'{255:X}'
('0xff', 'ff', 'FF')
```

See also `format()` for more information.

See also `int()` for converting a hexadecimal string to an integer using a base of 16.

> **Note:** To obtain a hexadecimal string representation for a float, use the `float.hex()` method.

id(*object, /*)

Return the "identity" of an object. This is an integer which is guaranteed to be unique and constant for this object during its lifetime. Two objects with non-overlapping lifetimes may have the same `id()` value.

CPython implementation detail: This is the address of the object in memory.

Raises an auditing event `builtins.id` with argument `id`.

input()
input(*prompt, /*)

If the *prompt* argument is present, it is written to standard output without a trailing newline. The function then reads a line from input, converts it to a string (stripping a trailing newline), and returns that. When EOF is read, `EOFError` is raised. Example:

```
>>> s = input('--> ')
--> Monty Python's Flying Circus
>>> s
"Monty Python's Flying Circus"
```

If the `readline` module was loaded, then `input()` will use it to provide elaborate line editing and history features.

Raises an auditing event `builtins.input` with argument `prompt` before reading input

Raises an auditing event `builtins.input/result` with the result after successfully reading input.

class **int**(*number=0, /*)
class **int**(*string, /, base=10*)

Return an integer object constructed from a number or a string, or return `0` if no arguments are given.

Examples:

```
>>> int(123.45)
123
>>> int('123')
123
>>> int('   -12_345\n')
-12345
>>> int('FACE', 16)
64206
>>> int('0xface', 0)
64206
>>> int('01110011', base=2)
115
```

If the argument defines `__int__()`, `int(x)` returns `x.__int__()`. If the argument defines `__index__()`, it returns `x.__index__()`. For floating-point numbers, this truncates towards zero.

If the argument is not a number or if *base* is given, then it must be a string, `bytes`, or `bytearray` instance representing an integer in radix *base*. Optionally, the string can be preceded by `+` or `-` (with no space in between), have leading zeros, be surrounded by whitespace, and have single underscores interspersed between digits.

A base-n integer string contains digits, each representing a value from 0 to n-1. The values 0–9 can be represented by any Unicode decimal digit. The values 10–35 can be represented by `a` to `z` (or `A` to `Z`). The default *base* is 10. The allowed bases are 0 and 2–36. Base-2, -8, and -16 strings can be optionally prefixed with `0b`/`0B`, `0o`/`0O`, or `0x`/`0X`, as with integer literals in code. For base 0, the string is interpreted in a similar way to an integer literal in code, in that the actual base is 2, 8, 10, or 16 as determined by the prefix. Base 0 also disallows leading zeros: `int('010', 0)` is not legal, while `int('010')` and `int('010', 8)` are.

The integer type is described in Numeric Types — int, float, complex.

Changed in version 3.4: If *base* is not an instance of `int` and the *base* object has a `base.__index__` method, that method is called to obtain an integer for the base. Previous versions used `base.__int__` instead of `base.__index__`.

Changed in version 3.6: Grouping digits with underscores as in code literals is allowed.

Changed in version 3.7: The first parameter is now positional-only.

Changed in version 3.8: Falls back to `__index__()` if `__int__()` is not defined.

Changed in version 3.11: `int` string inputs and string representations can be limited to help avoid denial of service attacks. A `ValueError` is raised when the limit is exceeded while converting a string to an `int` or when converting an `int` into a string would exceed the limit. See the integer string conversion length limitation documentation.

Changed in version 3.14: `int()` no longer delegates to the `__trunc__()` method.

isinstance(*object, classinfo, /*)

Return `True` if the *object* argument is an instance of the *classinfo* argument, or of a (direct, indirect, or virtual) subclass thereof. If *object* is not an object of the given type, the function always returns `False`. If *classinfo* is a tuple of type objects (or recursively, other such tuples) or a Union Type of multiple types, return `True` if *object* is an instance of any of the types. If *classinfo* is not a type or tuple of types and such tuples, a `TypeError` exception is raised. `TypeError` may not be raised for an invalid type if an earlier check succeeds.

Changed in version 3.10: *classinfo* can be a Union Type.

issubclass(*class, classinfo, /*)

Return `True` if *class* is a subclass (direct, indirect, or virtual) of *classinfo*. A class is considered a subclass of itself. *classinfo* may be a tuple of class objects (or recursively, other such tuples) or a Union Type, in which case return `True` if *class* is a subclass of any entry in *classinfo*. In any other case, a `TypeError` exception is raised.

Changed in version 3.10: *classinfo* can be a Union Type.

iter(*iterable, /*)
iter(*callable, sentinel, /*)

Return an iterator object. The first argument is interpreted very differently depending on the presence of the second argument. Without a second argument, the single argument must be a collection object which supports the iterable protocol (the `__iter__()` method), or it must support the sequence protocol (the `__getitem__()` method with integer arguments starting at `0`). If it does not support either of those protocols, `TypeError` is raised. If the second argument, *sentinel*, is given, then the first argument must be a callable object. The iterator created in this case will call *callable* with no arguments for each call to its `__next__()` method; if the value returned is equal to *sentinel*, `StopIteration` will be raised, otherwise the value will be returned.

See also Iterator Types.

One useful application of the second form of `iter()` is to build a block-reader. For example, reading fixed-width blocks from a binary database file until the end of file is reached:

```
from functools import partial
with open('mydata.db', 'rb') as f:
    for block in iter(partial(f.read, 64), b''):
        process_block(block)
```

len(*object, /*)

Return the length (the number of items) of an object. The argument may be a sequence (such as a string, bytes, tuple, list, or range) or a collection (such as a dictionary, set, or frozen set).

CPython implementation detail: `len` raises `OverflowError` on lengths larger than `sys.maxsize`, such as `range(2 ** 100)`.

class **list**(*iterable=(), /*)

Rather than being a function, `list` is actually a mutable sequence type, as documented in Lists and Sequence Types — list, tuple, range.

locals()

Return a mapping object representing the current local symbol table, with variable names as the keys, and their currently bound references as the values.

At module scope, as well as when using `exec()` or `eval()` with a single namespace, this function returns the same namespace as `globals()`.

At class scope, it returns the namespace that will be passed to the metaclass constructor.

When using `exec()` or `eval()` with separate local and global arguments, it returns the local namespace passed in to the function call.

In all of the above cases, each call to `locals()` in a given frame of execution will return the *same* mapping object. Changes made through the mapping object returned from `locals()` will be visible as assigned, reassigned, or deleted local variables, and assigning, reassigning, or deleting local variables will immediately affect the contents of the returned mapping object.

In an optimized scope (including functions, generators, and coroutines), each call to `locals()` instead returns a fresh dictionary containing the current bindings of the function's local variables and any nonlocal cell references. In this case, name binding changes made via the returned dict are *not* written back to the corresponding local variables or nonlocal cell references, and assigning, reassigning, or deleting local variables and nonlocal cell references does *not* affect the contents of previously returned dictionaries.

Calling `locals()` as part of a comprehension in a function, generator, or coroutine is equivalent to calling it in the containing scope, except that the comprehension's initialised iteration variables will be included. In other scopes, it behaves as if the comprehension were running as a nested function.

Calling `locals()` as part of a generator expression is equivalent to calling it in a nested generator function.

Changed in version 3.12: The behaviour of `locals()` in a comprehension has been updated as described in **PEP 709**.

Changed in version 3.13: As part of **PEP 667**, the semantics of mutating the mapping objects returned from this function are now defined. The behavior in optimized scopes is now as described above. Aside from being defined, the behaviour in other scopes remains unchanged from previous versions.

map(*function, iterable, /, *iterables, strict=False*)

Return an iterator that applies *function* to every item of *iterable*, yielding the results. If additional *iterables* arguments are passed, *function* must take that many arguments and is applied to the items from all iterables in parallel. With multiple iterables, the iterator stops when the shortest iterable is exhausted. If *strict* is `True` and one of the iterables is exhausted before the others, a `ValueError` is raised. For cases where the function inputs are already arranged into argument tuples, see `itertools.starmap()`.

Changed in version 3.14: Added the *strict* parameter.

max(*iterable, /, *, key=None*)
max(*iterable, /, *, default, key=None*)
max(*arg1, arg2, /, *args, key=None*)

Return the largest item in an iterable or the largest of two or more arguments.

If one positional argument is provided, it should be an iterable. The largest item in the iterable is returned. If two or more positional arguments are provided, the largest of the positional arguments is returned.

There are two optional keyword-only arguments. The *key* argument specifies a one-argument ordering function like that used for `list.sort()`. The *default* argument specifies an object to return if the provided iterable is empty. If the iterable is empty and *default* is not provided, a `ValueError` is raised.

If multiple items are maximal, the function returns the first one encountered. This is consistent with other sort-stability preserving tools such as `sorted(iterable, key=keyfunc, reverse=True)[0]` and `heapq.nlargest(1, iterable, key=keyfunc)`.

Changed in version 3.4: Added the *default* keyword-only parameter.

Changed in version 3.8: The *key* can be `None`.

class **memoryview**(*object*)

Return a "memory view" object created from the given argument. See Memory Views for more information.

min(*iterable, /, *, key=None*)
min(*iterable, /, *, default, key=None*)
min(*arg1, arg2, /, *args, key=None*)

Return the smallest item in an iterable or the smallest of two or more arguments.

If one positional argument is provided, it should be an iterable. The smallest item in the iterable is returned. If two or more positional arguments are provided, the smallest of the positional arguments is returned.

There are two optional keyword-only arguments. The *key* argument specifies a one-argument ordering function like that used for `list.sort()`. The *default* argument specifies an object to return if the provided iterable is empty. If the iterable is empty and *default* is not provided, a `ValueError` is raised.

If multiple items are minimal, the function returns the first one encountered. This is consistent with other sort-stability preserving tools such as `sorted(iterable, key=keyfunc)[0]` and `heapq.nsmallest(1, iterable, key=keyfunc)`.

Changed in version 3.4: Added the *default* keyword-only parameter.

Changed in version 3.8: The *key* can be `None`.

next(*iterator, /*)
next(*iterator, default, /*)

Retrieve the next item from the iterator by calling its `__next__()` method. If *default* is given, it is returned if the iterator is exhausted, otherwise `StopIteration` is raised.

class **object**

This is the ultimate base class of all other classes. It has methods that are common to all instances of Python classes. When the constructor is called, it returns a new featureless object. The constructor does not accept any arguments.

> **Note:** `object` instances do *not* have `__dict__` attributes, so you can't assign arbitrary attributes to an instance of `object`.

oct(*integer, /*)

Convert an integer number to an octal string prefixed with "0o". The result is a valid Python expression. If *integer* is not a Python `int` object, it has to define an `__index__()` method that returns an integer. For example:

```
>>> oct(8)
'0o10'
>>> oct(-56)
'-0o70'
```

If you want to convert an integer number to an octal string either with the prefix "0o" or not, you can use either of the following ways.

```
>>> '%#o' % 10, '%o' % 10
('0o12', '12')
>>> format(10, '#o'), format(10, 'o')
('0o12', '12')
>>> f'{10:#o}', f'{10:o}'
('0o12', '12')
```

See also `format()` for more information.

open(*file, mode='r', buffering=-1, encoding=None, errors=None, newline=None, closefd=True, opener=None*)

Open *file* and return a corresponding file object. If the file cannot be opened, an `OSError` is raised. See Reading and Writing Files for more examples of how to use this function.

file is a path-like object giving the pathname (absolute or relative to the current working directory) of the file to be opened or an integer file descriptor of the file to be wrapped. (If a file descriptor is given, it is closed when the returned I/O object is closed unless *closefd* is set to `False`.)

mode is an optional string that specifies the mode in which the file is opened. It defaults to `'r'` which means open for reading in text mode. Other common values are `'w'` for writing (truncating the file if it already exists), `'x'` for exclusive creation, and `'a'` for appending (which on *some* Unix systems, means that *all* writes append to the end of the file regardless of the current seek position). In text mode, if *encoding* is not specified the encoding used is platform-dependent: `locale.getencoding()` is called to get the current locale encoding. (For reading and writing raw bytes use binary mode and leave *encoding* unspecified.) The available modes are:

Character	Meaning
`'r'`	open for reading (default)
`'w'`	open for writing, truncating the file first
`'x'`	open for exclusive creation, failing if the file already exists
`'a'`	open for writing, appending to the end of file if it exists
`'b'`	binary mode
`'t'`	text mode (default)
`'+'`	open for updating (reading and writing)

The default mode is `'r'` (open for reading text, a synonym of `'rt'`). Modes `'w+'` and `'w+b'` open and truncate the file. Modes `'r+'` and `'r+b'` open the file with no truncation.

As mentioned in the Overview, Python distinguishes between binary and text I/O. Files opened in binary mode (including `'b'` in the *mode* argument) return contents as `bytes` objects without any decoding. In text mode (the default, or when `'t'` is included in the *mode* argument), the contents of the file are returned as `str`, the bytes having been first decoded using a platform-dependent encoding or using the specified *encoding* if given.

> **Note:** Python doesn't depend on the underlying operating system's notion of text files; all the processing is done by Python itself, and is therefore platform-independent.

buffering is an optional integer used to set the buffering policy. Pass 0 to switch buffering off (only allowed in binary mode), 1 to select line buffering (only usable when writing in text mode), and an integer > 1 to indicate the size in bytes of a fixed-size chunk buffer. Note that specifying a buffer size this way applies for binary buffered I/O, but `TextIOWrapper` (i.e., files opened with `mode='r+'`) would have another buffering. To disable buffering in `TextIOWrapper`, consider using the `write_through` flag for `io.TextIOWrapper.reconfigure()`. When no *buffering* argument is given, the default buffering policy works as follows:

- Binary files are buffered in fixed-size chunks; the size of the buffer is `max(min(blocksize, 8 MiB), DEFAULT_BUFFER_SIZE)` when the device block size is available. On most systems, the buffer will typically be 128 kilobytes long.
- "Interactive" text files (files for which `isatty()` returns `True`) use line buffering. Other text files use the policy described above for binary files.

encoding is the name of the encoding used to decode or encode the file. This should only be used in text mode. The default encoding is platform dependent (whatever `locale.getencoding()` returns), but any text encoding supported by Python can be used. See the `codecs` module for the list of supported encodings.

errors is an optional string that specifies how encoding and decoding errors are to be handled—this cannot be used in binary mode. A variety of standard error handlers are available (listed under Error Handlers), though any error handling name that has been registered with `codecs.register_error()` is also valid. The standard names include:

- `'strict'` to raise a `ValueError` exception if there is an encoding error. The default value of `None` has the same effect.
- `'ignore'` ignores errors. Note that ignoring encoding errors can lead to data loss.
- `'replace'` causes a replacement marker (such as `'?'`) to be inserted where there is malformed data.
- `'surrogateescape'` will represent any incorrect bytes as low surrogate code units ranging from U+DC80 to U+DCFF. These surrogate code units will then be turned back into the same bytes when the `surrogateescape` error handler is used when writing data. This is useful for processing files in an unknown encoding.
- `'xmlcharrefreplace'` is only supported when writing to a file. Characters not supported by the encoding are replaced with the appropriate XML character reference `&#nnn;`.
- `'backslashreplace'` replaces malformed data by Python's backslashed escape sequences.
- `'namereplace'` (also only supported when writing) replaces unsupported characters with `\N{...}` escape sequences.

newline determines how to parse newline characters from the stream. It can be `None`, `''`, `'\n'`, `'\r'`, and `'\r\n'`. It works as follows:

- When reading input from the stream, if *newline* is `None`, universal newlines mode is enabled. Lines in the input can end in `'\n'`, `'\r'`, or `'\r\n'`, and these are translated into `'\n'` before being returned to the caller. If it is `''`, universal newlines mode is enabled, but line endings are returned to the caller untranslated. If it has any of the other legal values, input lines are only terminated by the given string, and the line ending is returned to the caller untranslated.
- When writing output to the stream, if *newline* is `None`, any `'\n'` characters written are translated to the system default line separator, `os.linesep`. If *newline* is `''` or `'\n'`, no translation takes place. If *newline* is any of the other legal values, any `'\n'` characters written are translated to the given string.

If *closefd* is `False` and a file descriptor rather than a filename was given, the underlying file descriptor will be kept open when the file is closed. If a filename is given *closefd* must be `True` (the default); otherwise, an error will be raised.

A custom opener can be used by passing a callable as *opener*. The underlying file descriptor for the file object is then obtained by calling *opener* with (*file*, *flags*). *opener* must return an open file descriptor (passing `os.open` as *opener* results in functionality similar to passing `None`).

The newly created file is non-inheritable.

The following example uses the dir_fd parameter of the `os.open()` function to open a file relative to a given directory:

```
>>> import os
>>> dir_fd = os.open('somedir', os.O_RDONLY)
>>> def opener(path, flags):
...     return os.open(path, flags, dir_fd=dir_fd)
...
>>> with open('spamspam.txt', 'w', opener=opener) as f:
...     print('This will be written to somedir/spamspam.txt', file=f)
...
>>> os.close(dir_fd)  # don't leak a file descriptor
```

The type of file object returned by the `open()` function depends on the mode. When `open()` is used to open a file in a text mode (`'w'`, `'r'`, `'wt'`, `'rt'`, etc.), it returns a subclass of `io.TextIOBase` (specifically `io.TextIOWrapper`). When used to open a file in a binary mode with buffering, the returned class is a subclass of `io.BufferedIOBase`. The exact class varies: in read binary mode, it returns an `io.BufferedReader`; in write binary and append binary modes, it returns an `io.BufferedWriter`, and in read/write mode, it returns an `io.BufferedRandom`. When buffering is disabled, the raw stream, a subclass of `io.RawIOBase`, `io.FileIO`, is returned.

See also the file handling modules, such as `fileinput`, `io` (where `open()` is declared), `os`, `os.path`, `tempfile`, and `shutil`.

Raises an auditing event `open` with arguments `path`, `mode`, `flags`.

The `mode` and `flags` arguments may have been modified or inferred from the original call.

> *Changed in version 3.3:*
> - The *opener* parameter was added.
> - The `'x'` mode was added.
> - `IOError` used to be raised, it is now an alias of `OSError`.
> - `FileExistsError` is now raised if the file opened in exclusive creation mode (`'x'`) already exists.

> *Changed in version 3.4:*
> - The file is now non-inheritable.

> *Changed in version 3.5:*
> - If the system call is interrupted and the signal handler does not raise an exception, the function now retries the system call instead of raising an `InterruptedError` exception (see **PEP 475** for the rationale).
> - The `'namereplace'` error handler was added.

> *Changed in version 3.6:*
> - Support added to accept objects implementing `os.PathLike`.
> - On Windows, opening a console buffer may return a subclass of `io.RawIOBase` other than `io.FileIO`.

> *Changed in version 3.11:* The `'U'` mode has been removed.

ord(*character*, /)

Return the ordinal value of a character.

If the argument is a one-character string, return the Unicode code point of that character. For example, `ord('a')` returns the integer `97` and `ord('€')` (Euro sign) returns `8364`. This is the inverse of `chr()`.

If the argument is a `bytes` or `bytearray` object of length 1, return its single byte value. For example, `ord(b'a')` returns the integer `97`.

pow(*base*, *exp*, *mod=None*)

Return *base* to the power *exp*; if *mod* is present, return *base* to the power *exp*, modulo *mod* (computed more efficiently than `pow(base, exp) % mod`). The two-argument form `pow(base, exp)` is equivalent to using the power operator: `base**exp`.

When arguments are builtin numeric types with mixed operand types, the coercion rules for binary arithmetic operators apply. For `int` operands, the result has the same type as the operands (after coercion) unless the second argument is negative; in that case, all arguments are converted to float and a float result is delivered. For example, `pow(10, 2)` returns `100`, but `pow(10, -2)` returns `0.01`. For a negative base of type `int` or `float` and a non-integral exponent, a complex result is delivered. For example, `pow(-9, 0.5)` returns a value close to `3j`. Whereas, for a negative base of type `int` or `float` with an integral exponent, a float result is delivered. For example, `pow(-9, 2.0)` returns `81.0`.

For `int` operands *base* and *exp*, if *mod* is present, *mod* must also be of integer type and *mod* must be nonzero. If *mod* is present and *exp* is negative, *base* must be relatively prime to *mod*. In that case, `pow(inv_base, -exp, mod)` is returned, where *inv_base* is an inverse to *base* modulo *mod*.

Here's an example of computing an inverse for `38` modulo `97`:

```
>>> pow(38, -1, mod=97)
23
>>> 23 * 38 % 97 == 1
True
```

> *Changed in version 3.8:* For `int` operands, the three-argument form of `pow` now allows the second argument to be negative, permitting computation of modular inverses.

> *Changed in version 3.8:* Allow keyword arguments. Formerly, only positional arguments were supported.

print(**objects*, *sep=' '*, *end='\n'*, *file=None*, *flush=False*)

Print *objects* to the text stream *file*, separated by *sep* and followed by *end*. *sep*, *end*, *file*, and *flush*, if present, must be given as keyword arguments.

All non-keyword arguments are converted to strings like `str()` does and written to the stream, separated by *sep* and followed by *end*. Both *sep* and *end* must be strings; they can also be `None`, which means to use the default values. If no *objects* are given, `print()` will just write *end*.

The *file* argument must be an object with a `write(string)` method; if it is not present or `None`, `sys.stdout` will be used. Since printed arguments are converted to text strings, `print()` cannot be used with binary mode file objects. For these, use `file.write(...)` instead.

Output buffering is usually determined by *file*. However, if *flush* is true, the stream is forcibly flushed.

> *Changed in version 3.3:* Added the *flush* keyword argument.

class **property**(*fget=None*, *fset=None*, *fdel=None*, *doc=None*)

Return a property attribute.

fget is a function for getting an attribute value. *fset* is a function for setting an attribute value. *fdel* is a function for deleting an attribute value. And *doc* creates a docstring for the attribute.

A typical use is to define a managed attribute `x`:

```
class C:
    def __init__(self):
        self._x = None

    def getx(self):
        return self._x

    def setx(self, value):
        self._x = value

    def delx(self):
        del self._x

    x = property(getx, setx, delx, "I'm the 'x' property.")
```

If *c* is an instance of *C*, `c.x` will invoke the getter, `c.x = value` will invoke the setter, and `del c.x` the deleter.

If given, *doc* will be the docstring of the property attribute. Otherwise, the property will copy *fget*'s docstring (if it exists). This makes it possible to create read-only properties easily using `property()` as a decorator:

```
class Parrot:
    def __init__(self):
        self._voltage = 100000

    @property
    def voltage(self):
        """Get the current voltage."""
        return self._voltage
```

The `@property` decorator turns the `voltage()` method into a "getter" for a read-only attribute with the same name, and it sets the docstring for *voltage* to "Get the current voltage."

@getter

@setter

@deleter

A property object has `getter`, `setter`, and `deleter` methods usable as decorators that create a copy of the property with the corresponding accessor function set to the decorated function. This is best explained with an example:

```
class C:
    def __init__(self):
        self._x = None

    @property
    def x(self):
        """I'm the 'x' property."""
        return self._x

    @x.setter
    def x(self, value):
        self._x = value

    @x.deleter
    def x(self):
        del self._x
```

This code is exactly equivalent to the first example. Be sure to give the additional functions the same name as the original property (`x` in this case.)

The returned property object also has the attributes `fget`, `fset`, and `fdel` corresponding to the constructor arguments.

> *Changed in version 3.5:* The docstrings of property objects are now writeable.

__name__

Attribute holding the name of the property. The name of the property can be changed at runtime.

> *Added in version 3.13.*

class **range**(*stop*, /)
class **range**(*start*, *stop*, *step=1*, /)

Rather than being a function, `range` is actually an immutable sequence type, as documented in Ranges and Sequence Types — list, tuple, range.

repr(*object*, /)

Return a string containing a printable representation of an object. For many types, this function makes an attempt to return a string that would yield an object with the same value when passed to `eval()`; otherwise, the representation is a string enclosed in angle brackets that contains the name of the type of the object together with additional information often including the name and address of the object. A class can control what this function returns for its instances by defining a `__repr__()` method. If `sys.displayhook()` is not accessible, this function will raise `RuntimeError`.

This class has a custom representation that can be evaluated:

```
class Person:
   def __init__(self, name, age):
      self.name = name
      self.age = age

   def __repr__(self):
      return f"Person('{self.name}', {self.age})"
```

reversed(*object*, /)

Return a reverse iterator. The argument must be an object which has a `__reversed__()` method or supports the sequence protocol (the `__len__()` method and the `__getitem__()` method with integer arguments starting at `0`).

round(*number*, *ndigits=None*)

Return *number* rounded to *ndigits* precision after the decimal point. If *ndigits* is omitted or is `None`, it returns the nearest integer to its input.

For the built-in types supporting `round()`, values are rounded to the closest multiple of 10 to the power minus *ndigits*; if two multiples are equally close, rounding is done toward the even choice (so, for example, both `round(0.5)` and `round(-0.5)` are `0`, and `round(1.5)` is `2`). Any integer value is valid for *ndigits* (positive, zero, or negative). The return value is an integer if *ndigits* is omitted or `None`. Otherwise, the return value has the same type as *number*.

For a general Python object `number`, `round` delegates to `number.__round__`.

> **Note:** The behavior of `round()` for floats can be surprising: for example, `round(2.675, 2)` gives `2.67` instead of the expected `2.68`. This is not a bug: it's a result of the fact that most decimal fractions can't be represented exactly as a float. See Floating-Point Arithmetic: Issues and Limitations for more information.

class **set**(*iterable=()*, /)

Return a new `set` object, optionally with elements taken from *iterable*. `set` is a built-in class. See also Set Types — set, frozenset for documentation about this class.

For other containers see the built-in `frozenset`, `list`, `tuple`, and `dict` classes, as well as the `collections` module.

setattr(*object*, *name*, *value*, /)

This is the counterpart of `getattr()`. The arguments are an object, a string, and an arbitrary value. The string may name an existing attribute or a new attribute. The function assigns the value to the attribute, provided the object allows it. For example, `setattr(x, 'foobar', 123)` is equivalent to `x.foobar = 123`.

name need not be a Python identifier as defined in Names (identifiers and keywords) unless the object chooses to enforce that, for example in a custom `__getattribute__()` or via `__slots__`. An attribute whose name is not an identifier will not be accessible using the dot notation, but is accessible through `getattr()` etc..

> **Note:** Since private name mangling happens at compilation time, one must manually mangle a private attribute's (attributes with two leading underscores) name in order to set it with `setattr()`.

class **slice**(*stop*, /)
class **slice**(*start*, *stop*, *step=None*, /)

Return a slice object representing the set of indices specified by `range(start, stop, step)`. The *start* and *step* arguments default to `None`.

Slice objects are also generated when slicing syntax is used. For example: `a[start:stop:step]` or `a[start:stop, i]`.

See `itertools.islice()` for an alternate version that returns an iterator.

start
stop
step

These read-only attributes are set to the argument values (or their default). They have no other explicit functionality; however, they are used by NumPy and other third-party packages.

> *Changed in version 3.12:* Slice objects are now hashable (provided `start`, `stop`, and `step` are hashable).

sorted(*iterable*, /, *, *key=None*, *reverse=False*)

Return a new sorted list from the items in *iterable*.

Has two optional arguments which must be specified as keyword arguments.

key specifies a function of one argument that is used to extract a comparison key from each element in *iterable* (for example, `key=str.lower`). The default value is `None` (compare the elements directly).

reverse is a boolean value. If set to `True`, then the list elements are sorted as if each comparison were reversed.

Use `functools.cmp_to_key()` to convert an old-style *cmp* function to a *key* function.

The built-in `sorted()` function is guaranteed to be stable. A sort is stable if it guarantees not to change the relative order of elements that compare equal — this is helpful for sorting in multiple passes (for example, sort by department, then by salary grade).

The sort algorithm uses only `<` comparisons between items. While defining an `__lt__()` method will suffice for sorting, **PEP 8** recommends that all six rich comparisons be implemented. This will help avoid bugs when using the same data with other ordering tools such as `max()` that rely on a different underlying method. Implementing all six comparisons also helps avoid confusion for mixed type comparisons which can call the reflected `__gt__()` method.

For sorting examples and a brief sorting tutorial, see Sorting Techniques.

@staticmethod

Transform a method into a static method.

A static method does not receive an implicit first argument. To declare a static method, use this idiom:

```
class C:
    @staticmethod
    def f(arg1, arg2, argN): ...
```

The `@staticmethod` form is a function decorator – see Function definitions for details.

A static method can be called either on the class (such as `C.f()`) or on an instance (such as `C().f()`). Moreover, the static method descriptor is also callable, so it can be used in the class definition (such as `f()`).

Static methods in Python are similar to those found in Java or C++. Also, see `classmethod()` for a variant that is useful for creating alternate class constructors.

Like all decorators, it is also possible to call `staticmethod` as a regular function and do something with its result. This is needed in some cases where you need a reference to a function from a class body and you want to avoid the automatic transformation to instance method. For these cases, use this idiom:

```
def regular_function():
    ...

class C:
    method = staticmethod(regular_function)
```

For more information on static methods, see The standard type hierarchy.

> *Changed in version 3.10:* Static methods now inherit the method attributes (`__module__`, `__name__`, `__qualname__`, `__doc__` and `__annotations__`), have a new `__wrapped__` attribute, and are now callable as regular functions.

class **str**(**, encoding='utf-8', errors='strict'*)
class **str**(*object*)
class **str**(*object, encoding, errors='strict'*)
class **str**(*object, *, errors*)

Return a `str` version of *object*. See `str()` for details.

`str` is the built-in string class. For general information about strings, see Text Sequence Type — str.

sum(*iterable*, /, *start=0*)

Sums *start* and the items of an *iterable* from left to right and returns the total. The *iterable*'s items are normally numbers, and the start value is not allowed to be a string.

For some use cases, there are good alternatives to `sum()`. The preferred, fast way to concatenate a sequence of strings is by calling `''.join(sequence)`. To add floating-point values with extended precision, see `math.fsum()`. To concatenate a series of iterables, consider using `itertools.chain()`.

> *Changed in version 3.8:* The *start* parameter can be specified as a keyword argument.

> *Changed in version 3.12:* Summation of floats switched to an algorithm that gives higher accuracy and better commutativity on most builds.

> *Changed in version 3.14:* Added specialization for summation of complexes, using same algorithm as for summation of floats.

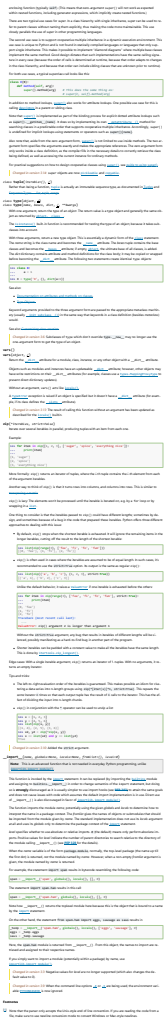
class **super**
class **super**(*type*, *object_or_type=None*, /)

Return a proxy object that delegates method calls to a parent or sibling class of *type*. This is useful for accessing inherited methods that have been overridden in a class.

The *object_or_type* determines the method resolution order to be searched. The search starts from the class right after the *type*.

For example, if `__mro__` of *object_or_type* is `D -> B -> C -> A -> object` and the value of *type* is `B`, then `super()` searches `C -> A -> object`.

The `__mro__` attribute of the class corresponding to *object_or_type* lists the method resolution search order used by both `getattr()` and `super()`. The attribute is dynamic and can change whenever the inheritance hierarchy is updated.

If the second argument is omitted, the super object returned is unbound. If the second argument is an object, `isinstance(obj, type)` must be true. If the second argument is a type, `issubclass(type2, type)` must be true (this is useful for classmethods).

When called directly within an ordinary method of a class, both arguments may be omitted ("zero-argument `super()`"). In this case, *type* will be the enclosing class, and *obj* will be the first argument of the immediately